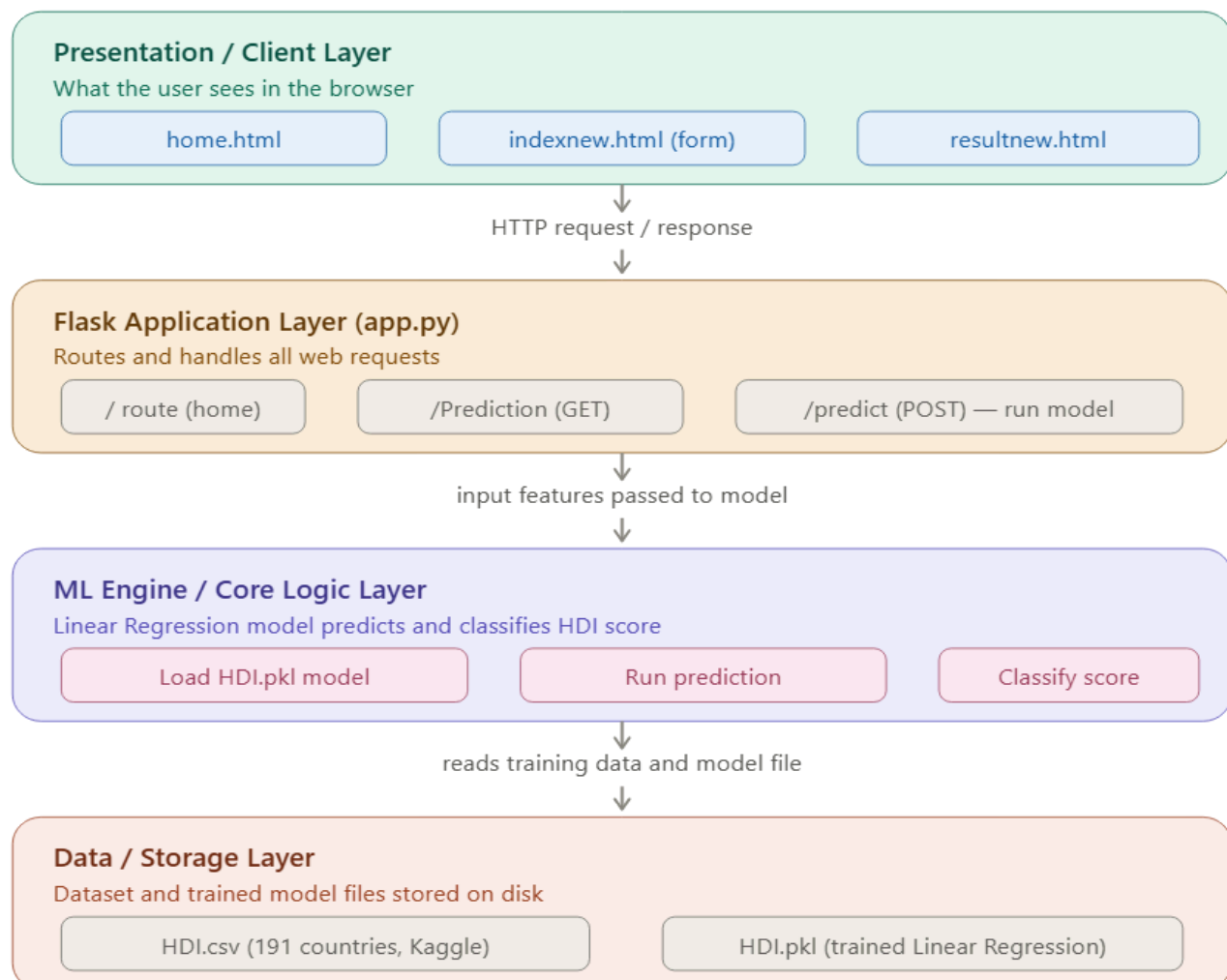


Solution Architecture

Date	29 June 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	The frontend user interface consisting of three web pages. home.html displays project introduction. indexnew.html shows the prediction input form with live HDI estimator. resultnew.html displays the predicted HDI score and category	HTML5, CSS3, JavaScript
API Gateway / Flask Router	Flask handles all routing between pages. Routes include / for home, /Prediction for the form page, and /predict (POST) for processing the prediction request and returning the result	Python 3.x, Flask Framework
Core Logic Service	The main ML prediction engine. Loads the saved HDI.pkl model, reads user inputs from the form, creates a Pandas DataFrame, runs Linear Regression prediction, classifies the score, and returns the result to the result page	scikit-learn Linear Regression, Pandas, NumPy, Pickle
Data / Storage Layer	HDI.csv stores the raw UNDP training dataset with 191 countries and 82 columns. HDI.pkl is the serialized trained Linear Regression model saved using Pickle for reuse without retraining	CSV file (Kaggle), Pickle (.pkl), Pandas DataFrame